ECE 310 Fall 2026
Lecture 13: Fourier Analysis

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Learning Objectives

- State what defines a continuous-time and discrete-time periodic function.
- Define and compute the continuous time Fourier series of a periodic continuous-time signal.*
- Understand conceptually how we may derive the continuous-time Fourier transform of aperiodic signals from the continuous-time Fourier series.
- Compute the discrete-time Fourier series of a periodic discrete-time signal.*
- Compute the discrete-time Fourier transform of a discrete-time signal or show that it does not exist.
- Explain what the discrete-time Fourier transform of a signal represents.



Periodic Signals: Continuous-time

Periodic Continuous-time Signals

A continuous-time signal $x(t)$ is said to be *periodic* with period T_0 (seconds) if

$$x(t_0) = x(t_0 + T_0).$$

- A given period T_0 has a corresponding linear frequency $F_0 = \frac{1}{T_0}$ (cycles/second or Hz).
- Radial frequency $\Omega_0 = 2\pi F_0$ (radians/second).
- The smallest possible T_0 for a given periodic signal is known as the *fundamental period* with a corresponding *fundamental frequency*.

Example periodic signals:

- $x(t) = \cos(\Omega_0 t)$
- $x(t) = e^{j\Omega_0 t}$



Periodic Signals: Discrete-time

Periodic Discrete-time Signals

A discrete-time signal $x[n]$ is said to be periodic with period N_0 (samples) if

$$x[n] = x[n + N_0].$$

- A given period N_0 has a corresponding linear frequency $f_0 = \frac{1}{N_0}$ (cycles/sample).
- Radial frequency $\omega_0 = 2\pi f_0$ (radians/sample).

Example periodic signals:

- $x[n] = \cos(\omega_0 n)$ where $\frac{\omega_0}{2\pi} = \frac{K}{L}$ for some integers K and L.
- $x[n] = e^{j\omega_0 n}$ where $\frac{\omega_0}{2\pi} = \frac{K}{L}$ for some integers K and L.



What frequencies are possible?

Continuous-time signals, e.g. $x(t) = e^{j\Omega_0 t}$:

$$\Omega_0 \in \mathbb{R} \rightarrow \quad -\infty < \Omega_0 < \infty$$

Discrete-time signals, e.g. $x[n] = e^{j\omega_0 n}$:

Claim: Periodic discrete-time signals only have unique frequencies between $[0, 2\pi)$ for ω_0.

Proof: Let $x[n] = e^{j(\omega_0 + 2\pi k)n}$, integer k

$$= e^{j\omega_0 n} e^{j2\pi kn}$$

$$= e^{j\omega_0 n} e^{j2\pi m}, \quad m = kn \in \mathbb{Z} \text{ (integer)}$$

$$= e^{j\omega_0 n}$$

Concept Check: Periodicity of Digital Signals

Let $x[n] = \sin\left(\frac{\pi}{3}n\right)$. Which of the following signals are equivalent to $x[n]$? **(Select all that apply)**

(a) $\sin\left(-\frac{\pi}{3}n\right)$

(b) $\sin\left(-\frac{5\pi}{3}n\right)$

(c) $\sin\left(\frac{4\pi}{3}n\right)$

(d) $\sin\left(\frac{13\pi}{3}n\right)$

$\sin\left(\frac{\pi}{3}n\right) \equiv \sin\left(\left(\frac{\pi}{3}+2\pi k\right)n\right)$

↑ equivalent to

$k=-1 \rightarrow \omega_0 = \frac{\pi}{3} - 2\pi = -\frac{5\pi}{3}$

$k=1 \rightarrow \omega_0 = \frac{\pi}{3} + 2\pi = \frac{7\pi}{3}$

$k=2 \rightarrow \omega_0 = \frac{\pi}{3} + 4\pi = \frac{13\pi}{3}$

Harmonically Related Signals

Two periodic signals are *harmonically related* if one signals has a frequency that is an integer multiple of the other. One common example is harmonically related complex exponential signals:

$$x_k(t) = e^{j\Omega_0 kt}, \ k \in \mathbb{Z}.$$

Key property of harmonically related complex exponentials: **Orthogonality Property**

$$\int_{t=0}^{T_0} x_k(t) x_l^*(t) = \begin{cases} 0, & k \neq l \\ T_0, & k = l \end{cases}$$

Proof is in Lecture 13 notes!



Continuous-time Fourier Series

Suppose we have a set of harmonically related complex exponential signals with fundamental frequency Ω_0, thus fundamental period $T_0 = \frac{2\pi}{\Omega_0}$. We can synthesize new periodic signals as a linear combination of these harmonically related signals:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\Omega_0 t}$$

What does each c_k represent?

c_k represents how much frequency $k\Omega_0$ is in $x(t)$



Continuous-time Fourier Series

Suppose we are given a periodic signal $x(t)$ with period $T_0 = \frac{2\pi}{\Omega_0}$. How can we determine the corresponding c_k coefficients?

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\Omega_0 k t}$$



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$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\Omega_0 kt}$$

$$\int_{T_0} x(t) e^{-j\Omega_0 lt} dt = \int_{T_0} \left(\sum_{k=-\infty}^{\infty} c_k e^{j\Omega_0 kt} \right) e^{-j\Omega_0 lt} dt$$



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Orthogonality property

$$\int_{T_0} x(t)e^{-j\Omega_0 lt}dt = \begin{cases} 0, & k \neq l \\ c_k T_0, & k = l \end{cases}$$

$$c_k = \frac{1}{T_0} \int_{T_0} x(t)e^{-j\Omega_0 kt}dt$$

Continuous-time Fourier Series Example

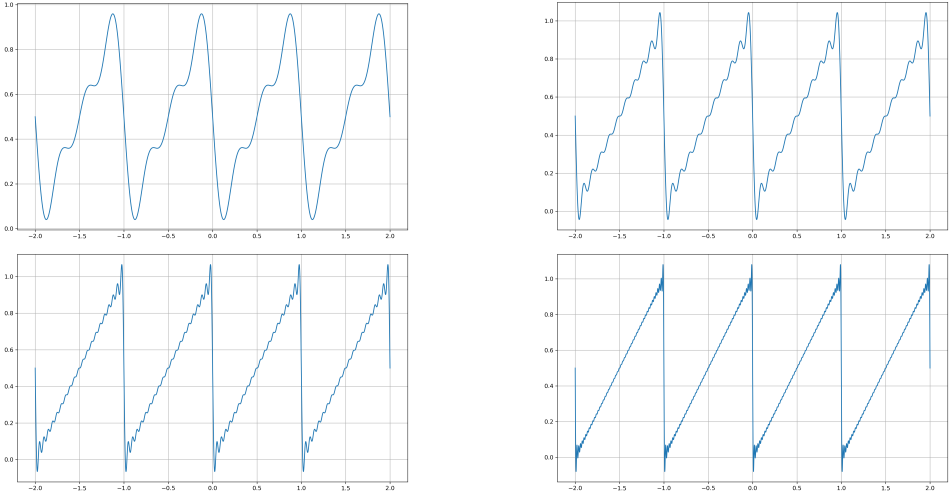


Figure 1: Partial sums for the Fourier series of a sawtooth wave $x(t) = t$ with period $T_0 = 1$. The Fourier series is given by $x(t) = \sum_{k=-m}^{m} c_k e^{j2\pi kt}$ with $c_k = \frac{j}{\pi k}$ for $k \neq 0$ and $c_0 = \frac{1}{2}$.

Continuous-time Fourier Series of Aperiodic Signals

What if we would like to decompose aperiodic signals?



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Aperiodic signals are the limit of periodic signals as $T_0 \to \infty$ and $\Omega_0 \to 0$.



Continuous-time Fourier Series of Aperiodic Signals

What if we would like to decompose aperiodic signals?

Aperiodic signals are the limit of periodic signals as $T_0 \to \infty$ and $\Omega_0 \to 0$. Informally, if we take the limit at T_0 for the continuous-time Fourier series,

$$c_k = \frac{1}{T_0} \int_{T_0} x(t) e^{-j\Omega_0 k t} dt \qquad \xrightarrow{T \to \infty,\ \Omega_0 \to 0} \qquad X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{j\Omega_0 k t} \qquad\qquad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$



Continuous-time Fourier Transform

The continuous-time Fourier transform (CTFT) of the signal $x(t)$ is denoted by $X(\Omega)$. Similarly, we often write $x(t) \overset{\mathcal{F}}{\longleftrightarrow} X(\Omega)$.

$$X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega$$

The CTFT of signal $x(t)$ exists if

$$\int_{-\infty}^{\infty} |x(t)|^2 dt < \infty.$$



Discrete-time Fourier Series

Similar to continuous-time signals, we can define harmonically related complex exponentials with frequency ω_0 and period $N_0 = \frac{2\pi}{\omega_0}$ as

$$x_k[n] = e^{j\omega_0 kn},\ 0 \leq k \leq N_0 - 1.$$



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$$x_k[n] = e^{j\omega_0 kn},\ 0 \le k \le N_0 - 1.$$

We may then define the discrete-time Fourier series (DTFS) as

$$x[n] = \sum_{k=0}^{N_0-1} c_k e^{j\frac{2\pi k}{N_0}n}$$

$$c_k = \frac{1}{N_0}\sum_{n=0}^{N_0-1} x[n]e^{-j\frac{2\pi k}{N_0}n},\ 0 \le k \le N_0 - 1$$



Discrete-time Fourier Series

Similar to continuous-time signals, we can define harmonically related complex exponentials with frequency ω_0 and period $N_0 = \frac{2\pi}{\omega_0}$ as

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$$x[n] = \sum_{k=0}^{N_0-1} c_k e^{j\frac{2\pi k}{N_0}n}$$

$$c_k = \frac{1}{N_0} \sum_{n=0}^{N_0-1} x[n] e^{-j\frac{2\pi k}{N_0}n},\ 0 \leq k \leq N_0 - 1$$

Similar to how we arrived at the CTFT, we may take the limit as $N_0 \to \infty$ and $\omega_0 \to 0$.



Discrete-time Fourier Transform

The discrete-time Fourier transform (DTFT) of a discrete-time signal $x[n]$ is denoted as $X_d(\omega)$: $x[n] \overset{\mathcal{F}}{\longleftrightarrow} X_d(\omega)$.

$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$x[n] = \frac{1}{2\pi} \int_{2\pi} X_d(\omega) e^{j\omega n} d\omega$$

The DTFT of the signal $x[n]$ exists if

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty.$$



Example: Computing the DTFT

Determine the DTFT of $x[n] = \{\underset{\uparrow}{1}, 0, 1, 0, 1\}$

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$$X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} (\delta[n] + \delta[n-2] + \delta[n-4]) e^{-j\omega n}$$

$$= (1)e^{-j\omega(0)} + (1)e^{-j\omega(2)} + (1)e^{-j\omega(4)}$$

$$= 1 + e^{-j2\omega} + e^{-j4\omega}$$

$$= e^{-j2\omega}\left(e^{j2\omega} + 1 + e^{-j2\omega}\right)$$

$$= e^{-j2\omega}\left(e^{j2\omega} + e^{-j2\omega} + 1\right)$$

$$= e^{-j2\omega}\left(2\cos(2\omega) + 1\right)$$

$$\cos(\theta) = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

Concept Check: Computing the DTFT

The DTFT of a discrete-time signal $x[n]$ is given by $X_d(\omega) = 2j\sin(\omega)$. What is $x[n]$?

Recall that $X_d(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$.

(a) $x[n] = \{\underset{\uparrow}{1}, 0, -1\}$ → $X_d(\omega) = 1 - e^{-j2\omega}$

(b) $x[n] = \{1, \underset{\uparrow}{0}, -1\}$ → $X_d(\omega) = e^{j\omega} - e^{-j\omega}$

(c) $x[n] = \{\underset{\uparrow}{2}, 0, -2\}$ → $X_d(\omega) = 2 - e^{-j2\omega}$

(d) $x[n] = \{2, \underset{\uparrow}{0}, -2\}$ → $X_d(\omega) = 2e^{j\omega} - 2e^{-j\omega}$